BILIRUBIN AND JAUNDICE

DEGRADATION OF HEME

- **LIFE SPAN OF ERYTHROCYTES IS 120 DAYS.**
- **DEGRADED BY MACROPHAGES OF THE RE SYSTEM IN THE SPLEEN AND LIVER.**
- **6 GM OF HEMOGLOBIN DEGRADED AND RESYNTHESIZED PER DAY.**
- **HEMOGLOBIN-----→ HEME +GLOBIN**

1.FORMATION OF BILIRUBIN

- **(CYCLIC)HEME CONVERTS IN TO (LINEAR) BILIVERDIN.**
- **CATALYZED BY HEME OXYGENASE SYSTEM.**
- **UTILIZES NADPH AND O_2.**
- **OPENING OF PORPHYRIN RINGS BY THREE SUCCESSIVE OXYGENATIONS.**
- **CO AND Fe^{2+} ARE PRODUCED.**

- **BILIVERDIN 'S METHENYL BRIDGES ARE REDUCED TO METHYLENE GROUP TO FORM BILIRUBIN.**

- **CATALYSED BY NADPH DEPENDENT BILIVERDIN REDUCTASE.**

- **1 GM OF HEMOGLOBIN YIELDS 35 mg OF BILIRUBIN.**

TRANSPORT OF BILIRUBIN TO LIVER

- **BILIRUBIN BINDS NONCOVENTLY WITH ALBUMIN.**
- **ALBUMIN-BILIRUBIN COMPLEX ENTERS THE LIVER BY FACILITATED DIFFUSION.**
- **BINDS TO INTRACELLULAR PROTEIN LIGANDIN.**

FORMATION OF DIGLUCURONIDE

- IN HEPATOCYTES BILIRUBIN IS CONGUGATED WITH TWO MOLECULES OF UDP-GLUCURONIC ACID .

- BILIRUBIN DIGLUCURONIDE IS FORMED.

- CATALYZED BY MICROSOMAL BILIRUBIN GLUCURONYLTRANSFERASE.
- LIVER CONJUGATE ABOUT 3.0g OF BILIRUBIN PER DAY

SECRETION OF BILIRUBIN INTO BILE

- CONJUGATED BILIRUBIN IS EXCRETED AGAINST THE CONCENTRATION GRADIENT, ENTERS INTO BILE
- TRANSPORT OF BILIRUBIN DIGLUCURONIDE IS AN ACTIVE, ENERGY- DEPENDENT AND RATE LIMITING PROCESS
- ALL BILIRUBIN (>98%) THAT ENTERS BILE IS IN THE CONJUGATED FORM

FATE OF BILIRUBIN

- HYDROLYSED IN THE INTESTINE BY SPECIFIC ENZYMES NAMELY β-GLUCURONIDASES TO LIBERATE BILIRUBIN
- CONVERTED TO UROBILINOGEN
- UROBILINOGEN CAN BE CONVERTED TO UROBILIN
- UROBILIN IS EXCRETED BY KIDNEY IN URINE
- UROBILINOGEN IS OXIDIZED BY BACTERIA TO STERCOBILIN WHICH IS EXCRETED WITH FECES

Formation of Bilirubin from Heme

Heme is degraded **in RE system (esp. liver & spleen)**
85% from RBCs
15% from turnover of immature RBCs & cytochromes

Heme

⇩ heme oxygenase

biliverdin (green)

⇩

bilirubin (red-orange)
bile pigments

⇩

In Blood with albumin
UNCONJUGATED BILIRUBIN
(or INDIRECT BILITUBIN)

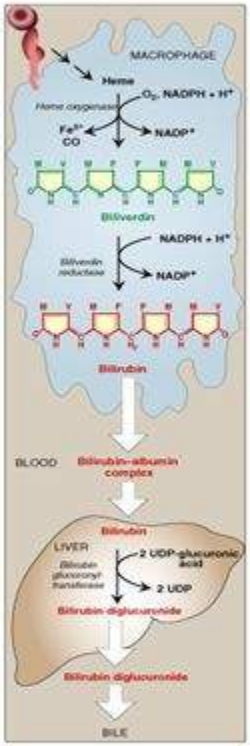


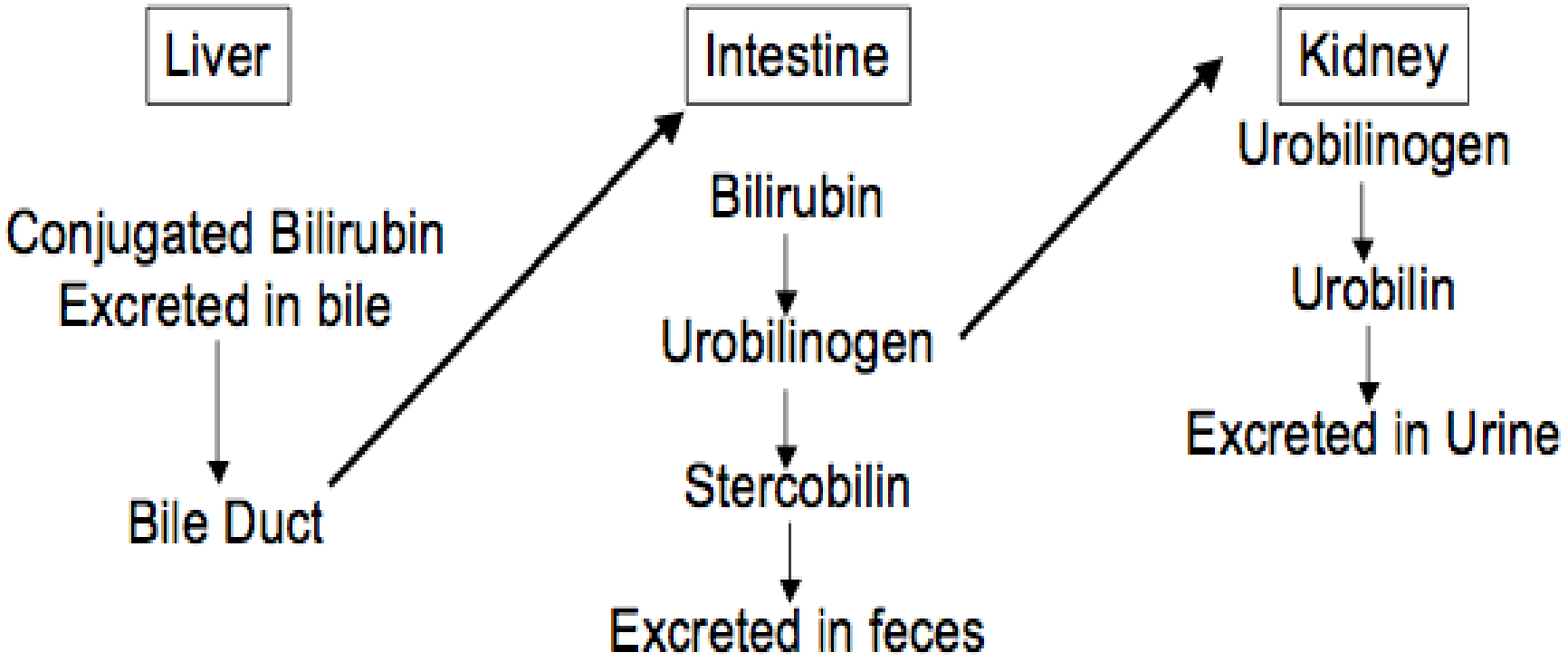
Figure 21.8
Formation of bilirubin from heme.

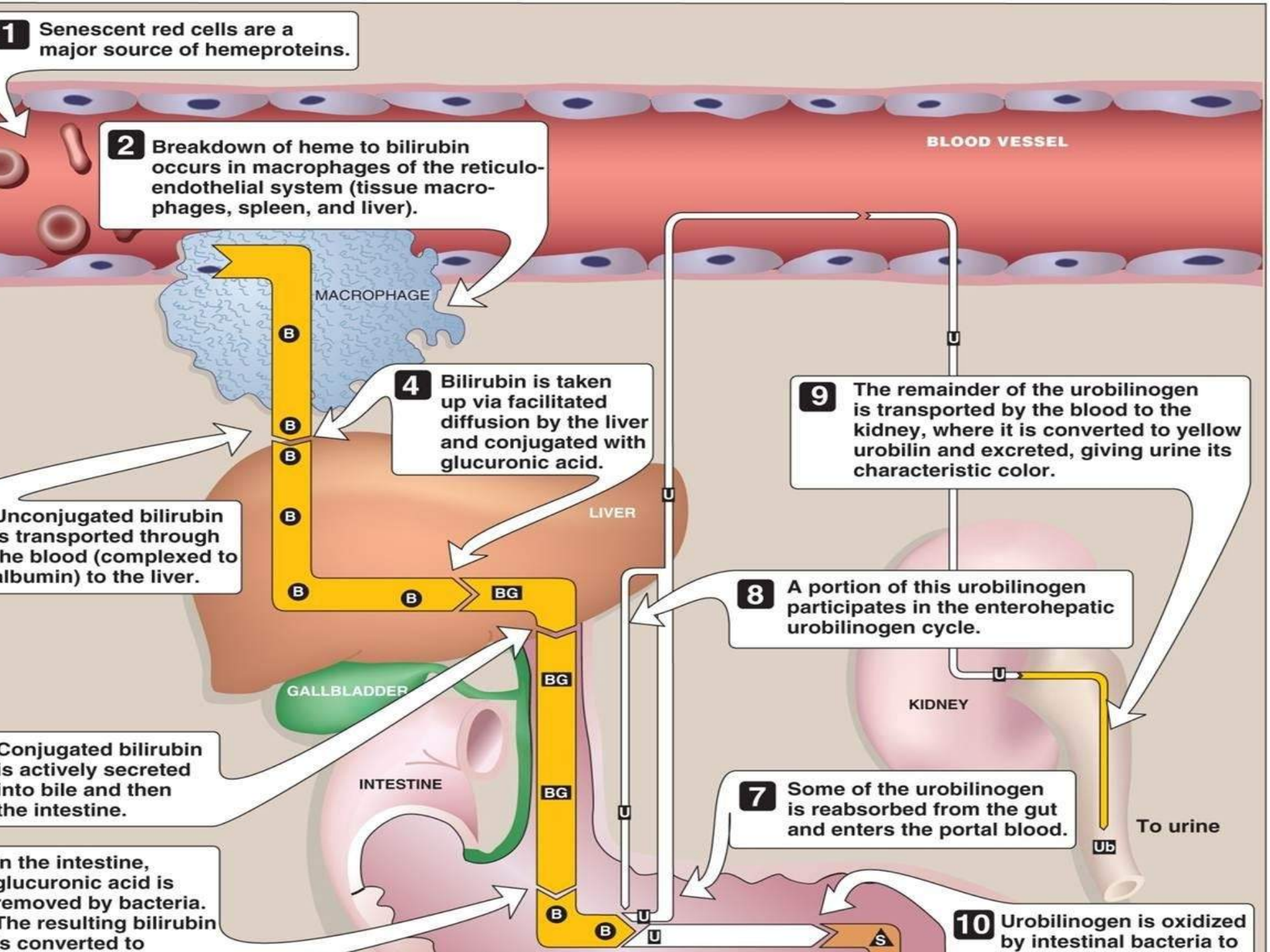
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Salicylates & sulfonamides can displace bilirubin from albumin & so bilirubin enters CNS causing **neural damage**

Bilirubin Excretion

Excreted actively by the liver in bile




1 Senescent red cells are a major source of hemeproteins.
2 Breakdown of heme to bilirubin occurs in macrophages of the reticulo-endothelial system (tissue macrophages, spleen, and liver).
BLOOD VESSEL
MACROPHAGE
4 Bilirubin is taken up via facilitated diffusion by the liver and conjugated with glucuronic acid.
Unconjugated bilirubin is transported through the blood (complexed to albumin) to the liver.
LIVER
9 The remainder of the urobilinogen is transported by the blood to the kidney, where it is converted to yellow urobilin and excreted, giving urine its characteristic color.
8 A portion of this urobilinogen participates in the enterohepatic urobilinogen cycle.
GALLBLADDER
KIDNEY
Conjugated bilirubin is actively secreted into bile and then the intestine.
INTESTINE
7 Some of the urobilinogen is reabsorbed from the gut and enters the portal blood.
To urine
In the intestine, glucuronic acid is removed by bacteria. The resulting bilirubin is converted to
10 Urobilinogen is oxidized by intestinal bacteria to
B
BG
U
S
Ub

BILIRUBIN

NORMAL

- **TOTAL BILIRUBIN LEVEL = O.2mg/dl - 1.0mg/dl**

- **UNCONJUGATED BILIRUBIN=0.2mg/dl – 0.6mg/dl (INDIRECT BILIRUBIN)**

- **CONJUGATED BILIRUBIN =0.2mg/dl -0.4mg/dl (DIRECT)**

JAUNDICE

- JAUNDICE (FRENCH: JAUNE-YELLOW)
- CLINICAL CONDITION CHARACTERIZED BY YELLOW CLOUR OF THE SCLERAE AND SKIN
- ELEVATED LEVELS OF BILIRUBIN IN THE SERUM – HYPERBILIRUBINEMIA.

CLASSIFICATION OF JAUNDICE

- **JAUNDICE IS CLASSIFIED INTO THREE MAJOR TYPES :**

- **HEMOLYTIC (PREHEPATIC)**
- **HEPATIC**
- **OBSTRUCTIVE (POST HEPATIC)**

1. HEMOLYTIC JAUNDICE

- ASSOCIATED WITH INCREASED HEMOLYSIS OF ERYTHROCYTES (e.g: INCOMPATIBLE BLOOD TRANSFUSION, MALARIA, SICKLE-CELL ANEMIA OR PYRUVATE KINASE OR GLUCOSE -6-PHOSPHATE DEHYDROGENASE DEFICIENCY)
- IS CHARACTERIZED BY:

- SERUM UNCONJUGATED BILIRUBIN RAISED
- INCREASED UROBILINOGEN IN URINE
- DARK BROWN COLOUR OF FECES (STERCOBILINOGEN)

2. HEPATIC (HEPATOCELLULAR) JAUNDICE

- CAUSED BY DYSFUNCTION OF THE LIVER
- DUE TO VIRAL INFECTION, POISONS, TOXINS, CIRRHOSIS OF LIVER, CARDIAC FAILURE
- IS CHARACTERIZED BY:
  - UNCONJUGATED BILIRUBIN IS RAISED
  - INCREASED LEVEL OF (ALT) OR (SGPT) AND (SGOT) OR (AST)
  - PALE , CLAY COLOURED STOOLS
  - NAUSEA AND ANOREXIA EXPERIENCED

3. OBSTRUCTIVE JAUNDICE

- DUE TO AN OBSTRUCTION IN THE BILE DUCT, CAUSED BY GALLSTONES, TUMORS etc.
- RAISED CONJUGATED BILIRUBIN
- IS CHARACTERIZED BY:
  - RAISED CONJUGATED BILIRUBIN IN SERUM
  - SERUM ALKALINE PHOSPHATASE AND γ GT IS ELEVATED
  - DARK COLOUR URINE AND CLAY COLOURED FECES
  - STEATORRHEA.
  - NAUSEA AND GASTROINTESTINAL PAIN.

NEONATAL-PHYSIOLOGIC JAUNDICE

- CAUSED BY IMMATURE HEPATIC SYSTEM FOR THE UPTAKE, CONJUGATON AND SECRETION OF BILIRUBIN
- ACTIVITY OF THE ENZYME UDP-GLUCURONYLTRANSFERASE IS LOW
- UNCONJUGATED BILIRUBIN IS HIGHLY ELEVATED (MAY GO BEYOND 25mg/dl)
- RESULTS IN HYPERBILIRUBINEMIC TOXIC ENCEPHALOPATHY OR KERNICTERUS

PHOTOTHERAPY

- **BILIRUBIN CAN ABSORB BLUE LIGHT (420-470 nm) MAXIMALLY**
- **TOXIC NATIVE UNCONJUGATED BILIRUBIN GETS CONVERTED INTO A NON-TOXIC ISOMER NAMELY LUMIRUBIN**
- **LUMIRUBIN CAN BE EASILY EXCRETED BY THE KIDNEYS**

GENETIC DISORDERS

- **CRIGLER – NAJJAR SYNDROME TYPE I**
- **CRIGLER – NAJJAR SYNDROME TYPE II**
- **GILBERT'S DISEASE**

CRIGLER –NAJJAR SYNDROME TYPE I

- CONGENITAL NON-HEMOLYTIC JAUNDICE
- DEFECT IN THE HEPATIC ENZYME UDP-GLUCURONYLTRANSFERASE
- CHILDREN DIE WITHIN FIRST TWO YEARS OF LIFE

CRIGLER–NAJJAR SYNDROME TYPE II

- DUE TO A LESS SEVERE DEFECT IN THE BILIRUBIN CONJUGATION
- SERUM BILIRUBIN CONCENTRATION IS USUALLY LESS THAN 20 mg/dl AND IS LESS DANGEROUS

GILBERT'S DISEASE

- GILBERT'S DISEASE IS A COMBINATION OF DISORDERS
- THESE INCLUDE:
  - A DEFECT IN THE UPTAKE OF BILIRUBIN BY LIVER CELLS
  - AN IMPAIRMENT IN CONJUGATION DUE TO REDUCED ACTIVITY OF UDP-GLUCURONYLTRANSFERASE
  - DECREASED HEPATIC CLEARANCE OF BILIRUBIN